

Initial Info



OGN - Challenge 1 (solved by 89 teams)

60 ✓

Problem

[Previous Submissions](#)



After gaining admin access to the smart rifle system, what is the flag?

Please rate this problem: ☆☆☆☆☆



THE
FUTURE
IS*****

OGN - Challenge 1

You found a challenge!

Here's a small bonus for finding it: `KEY{f45dd8}`. Submit it as a "Bonus" [here](#).

Solve the challenge below for additional points. The flag will be in format `KEY{...}`:

Wheeler has gained access to the pirates' smart rifles via an interactive console. Help him take control of them!

Connect to the console with `nc challenges.tficommic.io 7005`.

(paste that command into your terminal or command prompt)

A copy of the smart rifle's program can be found [here](#).

Scoreboard 

Workflow

```

(venv)-(kali㉿kali)-[~/CTF_Files/the_future_is_..._comic]
$ nc challenges.tficommic.io 7005
===== Welcome to the USHKUYNIK smart rifle system! =====
What would you like to do?
1. Activate FRIEND-OR-FOE system
2. Enter administrator password
3. Exit
> 2

Please enter the administrator password: adim

ERROR: Incorrect administrator password
a
What would you like to do?
1. Activate FRIEND-OR-FOE system
2. Enter administrator password
3. Exit
>2

Please enter the administrator password: admin

ERROR: Incorrect administrator password

What would you like to do?
1. Activate FRIEND-OR-FOE system
2. Enter administrator password
3. Exit
> 1

ERROR: Administrator privileges required.

```

Statically analyzed the python code:

```

#!/usr/bin/python
import time

def main():
    flag = ''
    with open('./flag.txt', 'r') as f:
        flag = f.readline().strip()

    print("===== Welcome to the USHKUYNIK smart rifle system!
===== ", end="", flush=True)
    time.sleep(1)

    admin = False
    admin_pw_encrypted =
'6d6e71775b7174727b7d6f6f536460506479774c677877656c4668727a717b40494f465657515
45e'

```

```

while True:
    print("\nWhat would you like to do?\n", flush=True)
    print("1. Activate FRIEND-OR-FOE system")
    print("2. Enter administrator password")
    print("3. Exit")
    user_in = input('> ')

    if user_in == '1':
        if admin:
            print('\nFRIEND-OR-FOE system activated.', flush=True)
            time.sleep(0.5)
            print(f'\nCongratulations! Here is the flag: {flag}')
            exit()
        else:
            print('\nERROR: Administrator privileges required.',
flush=True)
            time.sleep(1)
    elif user_in == '2':
        print('\nPlease enter the administrator password: ', end="",
flush=True)
        pw = input("")
        admin_pw = ''.join(chr(ord(c) ^ i) for i, c in
enumerate(bytes.fromhex(admin_pw_encrypted).decode("utf-8"))))
        if pw == admin_pw:
            admin = True
            print('\nAdministrator password verified. Privileges
updated.', flush=True)
            time.sleep(1)
        else:
            print('\nERROR: Incorrect administrator password', flush=True)
            time.sleep(1)
    elif user_in == '3':
        exit()
    else:
        print('\nInvalid option. Please try again', flush=True)
        time.sleep(1)

if __name__ == "__main__":
    main()

```

Used the hard creds against it and just reversed the encoding algorithm in an online python compiler:

main.py	Output
<pre>1 admin_pw_encrypted = "6d6e71775b7174727b7d6f6f536460506479774c677877656c4668727a717b40494f46565751545e" 2 3 admin_pw = ''.join(4 chr(ord(c) ^ i) 5 for i, c in enumerate(bytes.fromhex(admin_pw_encrypted).decode("utf-8")) 6) 7 print(admin_pw) 8</pre>	<pre>most_trusted_in_the_smart_rifle_industry === Code Execution Successful ===</pre>

Flag Found after putting in the password:

```
What would you like to do?
1. Activate FRIEND-OR-FOE system
2. Enter administrator password
3. Exit
> 2

Please enter the administrator password: most_trusted_in_the_smart_rifle_industry
Administrator password verified. Privileges updated.

What would you like to do?
1. Activate FRIEND-OR-FOE system
2. Enter administrator password
3. Exit
> 1

FRIEND-OR-FOE system activated.

Congratulations! Here is the flag: KEY{4e3cae}
```